ALESSIO BRINI, PH.D.

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RESEARCH INTERESTS

Artificial intelligence and machine learning applied to finance and economics, with a focus on reinforcement learning for sequential decision problems in trading, hedging, and risk management. Financial econometrics and forecasting with data-driven and deep learning methods. Digital assets, decentralized finance, and the analysis of blockchain data.

EDUCATION

Ph.D. at Scuola Normale Superiore — 11/2018 - 09/2022
Mathematical Finance — *Pisa, Italy*

· Thesis: *Reinforcement learning for sequential decision-making: a data-driven approach for finance*

Master's degree at University of Florence — 2016 - 2018
Quantitative Finance and Risk Management — *Florence, Italy*

· Thesis: *Beyond the parameters' estimation burden: a Bayesian framework for option pricing*

Bachelor's degree at University of Florence — 2012 - 2015
Business and Economics — *Florence, Italy*

ACADEMIC APPOINTMENTS

Executive in Residence - Duke University - Pratt School of Engineering
07/2024 - present, *Durham, NC, USA*

· Leads the *Digital Asset Research & Engineering Collaborative (DAREC)* Lab.

· Teaches graduate courses in FinTech and leads research projects that apply machine learning and data science methods to problems in finance, particularly digital assets and DeFi.

Postdoctoral Researcher - Duke University - Pratt School of Engineering
01/2022 - 06/2024, *Durham, NC, USA*

· Applied machine learning and data science methods to problems in finance, with a focus on digital assets and DeFi. Worked with supervised, unsupervised, and reinforcement learning.

· Delivered coursework on *Machine Learning for FinTech*, *Data Wrangling and Visualization with Python*, and *Introduction to Statistics and Econometrics* for the Master of Engineering in Financial Technology program.

· Directed FinTech capstone projects spanning machine learning, data science, and digital ledger technologies.

· Led summer research projects on DeFi and machine learning for graduate students.

PUBLICATIONS

Peer-Reviewed Publications

1. "On Deep Reinforcement Learning for Dynamic Trading with PPO: Challenges and Future Directions" (with Petter Kolm). *The Journal of Financial Data Science*. 2026

2. "Beyond Average Hive Performance: Tail Risk Measurement in Italian Apiculture with Honey-at-Risk" (with Giacomo Toscano, Ginevra Virginia Lombardi, and Maria Elvira Mancino). *Environmetrics*. 2026

3. "StableAML: Machine Learning for Behavioral Wallet Detection in Stablecoin Anti-Money Laundering on Ethereum" (with Luciano Juvinski and Haochen Li). *Blockchain*. 2026

4. "Graph Neural Networks for Bridge Swap Link Prediction in Uniswap v3" (with Qingran Zhou and Eric Liu). *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF'25)*. 2025

5. "SpotV2Net: Multivariate Intraday Spot Volatility Forecasting via Vol-of-Vol-Informed Graph Attention Networks" (with Giacomo Toscano). *International Journal of Forecasting*. 2025

6. "Modeling and Simulating a Startup Ecosystem Development with a Delayed Differential Equation" (with Viviana Fanelli). *Soft Computing*. 2025

7. "Data-Driven Derivative Hedging with Quadratic Variation Penalty" (with Giacomo Domeniconi and Ali Fathi). *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF'24)*. 2024

8. "Pricing Cryptocurrency Options with Machine Learning Regression for Handling Market Volatility" (with Jimmie Lenz). *Economic Modelling*. 2024

9. "A Comparison of Cryptocurrency Volatility-Benchmarking New and Mature Asset Classes" (with Jimmie Lenz). *Financial Innovation*. 2024

10. "Reinforcement Learning Policy Recommendation for Interbank Network Stability" (with Gabriele Tedeschi and Daniele Tantari). *Journal of Financial Stability*. 2023

11. "Deep Reinforcement Trading with Predictable Returns" (with Daniele Tantari). *Physica A: Statistical Mechanics and its Applications*. 2023

12. "Assessing the Resiliency of Investors against Cryptocurrency Market Crashes through the Leverage Effect" (with Jimmie Lenz). *Economics Letters*. 2022

Working Papers

1. "A Nonlinear Target-Factor Model with Attention Mechanism for Mixed-Frequency Data" (with Ekaterina Seregina). arXiv:2601.16274. 2026

2. "Forecasting Realized Volatility with Time Series Foundation Models: A Comparison with Econometric Benchmarks". arXiv:2607.05291. 2026

3. "Empirical Models of the Time Evolution of SPX Option Prices" (with David Hsieh, Patrick Kuiper, Sean Moushegian, and David Ye). arXiv:2506.17511. 2025

4. "Improving DeFi Accessibility through Efficient Liquidity Provisioning with Deep Reinforcement Learning" (with Haonan Xu). arXiv:2501.07508. 2025

5. "A Machine Learning Approach to Forecasting Honey Production with Tree-Based Methods" (with Elisa Giovannini and Elia Smaniotto). arXiv:2304.01215. 2023

Other Articles and Lightly Refereed Publications

· "Crypto Dynamics during Market Downturns: Another Dotcom Boom-Bust Cycle?" (with Jimmie Lenz and Emma Rasiel). DAREC working paper. 2022

· "Bitcoin ETFs: Measuring the Performance of this New Market Niche" (with Jimmie Lenz). SSRN 4157711. 2022

GRANTS & FUNDING

· Member of the research consortium of the project PID2025-173261OB-I00, a computational simulator for green bank investment led by Gabriele Tedeschi at Universitat Jaume I, funded by the Spanish Ministry of Science, Innovation and Universities through the State Research Agency (Grant amount: 121k €).

· Co-Investigator on the project "Risk Management in Times of Unprecedented Geo-Political Volatility: A Machine-Learning Approach", funded by the Institut Europlace de Finance (IEF), Institut Louis Bachelier (Grant amount: 10k €).

· Collaborator in the "BEEkeepers Weather Indexed INsurance" project at the University of Florence, funded by the Italian Ministry of Agricultural, Food and Forestry Policies (Grant amount: 600k €).

CONFERENCE PRESENTATIONS & INVITED TALKS

Financial Econometrics Lunch Group Seminar Series - Duke University, Durham, NC, September 21, 2026:

· Talk: *A Nonlinear Target-Factor Model with Attention Mechanism for Mixed-Frequency Data.*

AIBThings 2026: IEEE 4th International Conference on Artificial Intelligence, Blockchain, and Internet of Things - Central Michigan University, Mount Pleasant, MI (online), September 5-6, 2026:

· Contributed Talk: *Do Rollups Self-Organize? Multi-Agent Reinforcement Learning in the EIP-4844 Blob Fee Market.*

Quaint Quant Conference - SMU Cox School of Business, Dallas, TX, April 10, 2026:

· Contributed Talk: *Attention-Based Mixed-Frequency Factor Models for Macroeconomic Forecasting.*

Conference on Society-Centered AI - Duke University, Durham, NC, February 12-14, 2026:

· Invited Talk, New Faculty Spotlight (video): *AI in Finance and Economics: From Data to Decisions that Matter.*

Monitoring Risks in Financial Markets Workshop - University of Macerata, Macerata, Italy, February 4, 2026:

· Invited Talk: *A Nonlinear Target-Factor Model with Attention Mechanism for Mixed-Frequency Data.*

Econometrics Seminar - University of California, Riverside, CA (online), January 6, 2026:

· Talk: *A Nonlinear Target-Factor Model with Attention Mechanism for Mixed-Frequency Data.* Presented by coauthor Ekaterina Seregina.

ICAIF'25: 6th ACM International Conference on AI in Finance - Singapore, November 15-18, 2025:

· Contributed Talk: *Graph Neural Networks for Bridge Swap Link Prediction in Uniswap v3.* Best student paper honorable mention (link).

Financial Econometrics Lunch Group Seminar Series - Duke University, Durham, NC, October 27, 2025:

· Talk: *A Nonlinear Target-Factor Model with Attention Mechanism for Mixed-Frequency Data.*

AAAI-25 Workshop on AI for Social Impact: Bridging Innovations in Finance, Social Media, and Crime Prevention - Philadelphia, PA, March 3, 2025:

· Contributed Talk: *Improving DeFi Accessibility through Efficient Liquidity Provisioning with Deep Reinforcement Learning.* Best paper award (link).

ICAIF'24: 5th ACM International Conference on AI in Finance - New York, NY, November 14-17, 2024:

· Poster Presentation: *Data-Driven Derivative Hedging with Quadratic Variation Penalty.*

2024 NBER-NSF Time Series Conference - Philadelphia, PA, September 20-21, 2024:

· Poster Presentation: *SpotV2Net: Multivariate Intraday Spot Volatility Forecasting via Vol-of-Vol-Informed Graph Attention Networks.*

CrunchDAO - Online, December 14, 2023:

· Invited Talk (video): *Reinforcement Learning for Sequential Decision-Making: A Data-Driven Approach.*

D2 Seminar Series, Florence Center for Data Science - University of Florence, Florence, Italy, October 28, 2022:

· Invited Talk: *Reinforcement Learning Policy Recommendation for Interbank Network Stability.*

Machine Learning of Dynamic Processes and Time Series Analysis - Pisa, Italy, November 26-27, 2020:

· Contributed Talk: *Trading Mean Reversion with Reinforcement Learning.*

13th European Financial Mathematics Summer School - Vienna, Austria, August 31 - September 4, 2020:

· Contributed Talk: *Trading Mean Reversion with Reinforcement Learning.*

Eastern European Machine Learning Summer School - Krakow, Poland (online), July 1-9, 2020:

· Poster Presentation: *Trading Mean Reversion with Reinforcement Learning.*

TEACHING EXPERIENCE

FINTECH 540: Machine Learning for FinTech - Pratt School of Engineering, Duke University:

Taught the theory and practice of machine learning in FinTech, with coding sessions on real financial data.

FINTECH/ECE 590: Data Wrangling and Visualization with Python - Pratt School of Engineering, Duke University:

Taught Python for querying, cleaning, manipulating, and visualizing data. Covered databases (MySQL and NoSQL), web scraping, and data collection.

FINTECH 502: FinTech Capstones - Pratt School of Engineering, Duke University:

Oversaw teams of students developing real-world technical solutions in the FinTech industry, with projects sponsored by industry professionals.

FINTECH 520: Introduction to Statistics and Econometrics - Pratt School of Engineering, Duke University:

Taught graduate FinTech students the statistical and econometric concepts and tools they use in later courses and in research.

Python for Data Science - University of Florence (Italy) - March-May and September-December 2020:

Taught the basics of the Python language and its main scientific packages for data analysis.

ADVISING & MENTORING

Graduate Students:

· Luciano Juvinski, Duke University, 2025: "A Scalable Framework for Stablecoin AML: Leveraging Supervised Learning Models for Behavioral Wallet Detection on Ethereum"

· Longyi Hu, Duke University, 2025: "Conformal Prediction for Finance and Economics"

· Qingran Zhou, Duke University, 2025: "Graph Neural Networks for Bridge Swap Link Detection on Uniswap"

· Haonan Xu, Duke University, 2024: "Improving DeFi Accessibility through Efficient Liquidity Provisioning with Deep Reinforcement Learning"

· Alex Ilgenfritz, Duke University, 2023: "Optimal Liquidity Provision in Uniswap v3 with Reinforcement Learning"

· Mike Liu, Duke University, 2022: Independent study on asset pricing and financial dashboard development

· Niccolò Tarchi, University of Florence, 2021: "Transformer-based approach for creating financial portfolios"

· Dmitry Guzairov, University of Florence, 2021: "Hierarchical risk-parity portfolios: a clustering approach to the investment problem"

· Duccio Guerrini, University of Florence, 2020: "Stock market indexes and machine learning algorithms. Replication of the S&P500 index by the autoencoder neural network"

PROFESSIONAL SERVICE

Editorial Service:

· Cogent Economics and Finance (Taylor & Francis), Associate Editor (2026-present)

Program Committee:

- ACM International Conference on AI in Finance (ICAIF) 2026, Program Committee Member and Meta-Reviewer
- IEEE International Conference on Big Data (IEEE BigData) 2026, Program Committee Member

Referee Service:

Journal of Financial Econometrics, Annals of Operations Research, Expert Systems with Applications, Financial Innovation, Finance Research Letters, Applied Economics Letters, Scientific Reports, Applied Economics, Journal of Applied Economics, Discover Artificial Intelligence, AI Computer Science and Robotics Technology, Data Science and Management, The Journal of Financial Data Science, Review of Derivatives Research, The B.E. Journal of Macroeconomics, International Journal of Data Science and Analytics, Journal of King Saud University Computer and Information Sciences, Journal of Evolutionary Economics, ACM International Conference on AI in Finance (ICAIF) 2024 and 2025, European Conference in AI for Finance (ECAIF) 2024 and 2025, IEEE International Conference on Machine Learning and Applications (ICMLA) 2026, IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings) 2026, International Conference on Applied Computational Intelligence and Systems Modeling (ACISM) 2026

Conference Organization:

- Organizer - Digital Assets at Duke (DA@D), 2023 and 2024, Durham, NC, USA
- Scientific Committee - Florence-Paris Workshop: Statistics of Random Processes and Its Applications to Financial Econometrics, July 2023, Florence, Italy

Departmental Service (Duke University, Pratt School of Engineering):

- Reviews applications and assists with recruitment for the FinTech graduate program (2023-present).
- Represents the program in outreach to industry and to the research community.